

Power and Propulsion Systems for UV

Exam Exercise 1 – Internal Combustion Engine

GT-Power (GT-SUITE) 1D Simulation Report

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Assigned scaling parameter	Value
Bore multiplier	$\times 0.97$
Stroke multiplier	$\times 1.04$
PX vehicle configuration (Part 2)	1
ICE torque map multiplier (Part 2)	$\times 1.02$

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1 Introduction and Model Setup

This report documents the four-part 1D engine-simulation exercise carried out in GT-Power (GT-SUITE), starting from the course template SI_4cyl_SITurb_PortFuelFilm_TWallSoln – a four-cylinder, port-fuel-injected, spark-ignition engine model featuring predictive SI combustion, a knock model, port fuel-film (puddling) dynamics, and an in-cylinder wall-temperature solver. The baseline geometry (bore, stroke, and intake/exhaust valve diameters) was individually scaled according to the matricola-specific multipliers listed on the title page: bore $\times 0.97$ and stroke $\times 1.04$ relative to the template engine, with valve diameters scaled proportionally to the new bore so that the breathing geometry remains self-consistent. The air–fuel ratio was held at the stoichiometric value for gasoline, $AFR_{stoich} \approx 14.7$, for the baseline case.

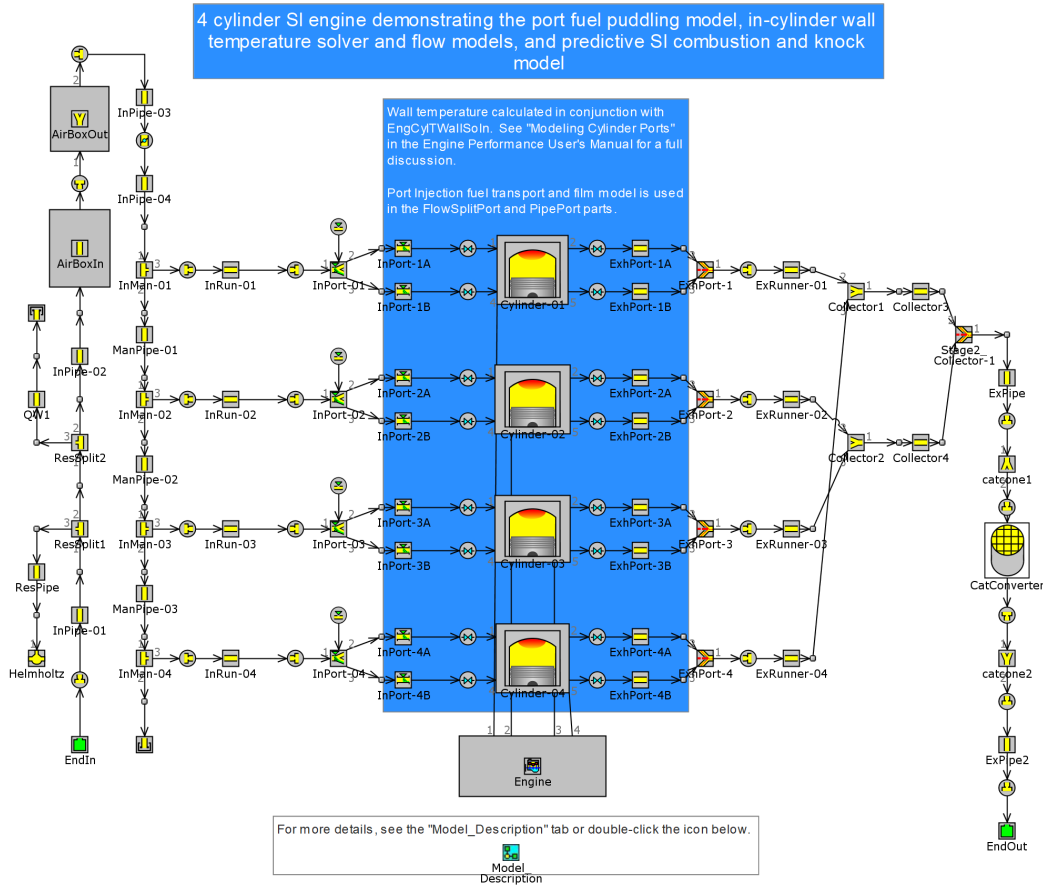


Figure 1: GT-Power schematic used for Cases 1–3: air box, four intake/exhaust runners, four cylinders (with wall-temperature solver), exhaust collectors, catalyist, and ambient boundary conditions.

Four sub-cases were investigated, each probing a different engine control/operating variable while holding the others fixed:

- Case 1. Engine speed sweep** – 1000, 2000, 3000, 4000, 5000 rpm at wide-open throttle and stoichiometric AFR.
- Case 2. Air–fuel ratio sweep** – the same speed sweep repeated at a rich mixture ($AFR = 11.76$, -20% vs. stoichiometric) and a lean mixture ($AFR = 17.64$, $+20\%$ vs. stoichiometric).
- Case 3. Spark advance sweep** – engine speed held fixed at 3000 rpm; spark timing varied across five settings, $-10, -15, -20, -25, -30$ °CA (degrees before top-dead-centre firing).

Case 4. Throttle position / load sweep – a closed-loop ThrottleController (Fig. 2) was added to target four fractions of maximum torque: 5%, 10%, 20%, and 50%, corresponding to throttle angles of 18°, 26°, 37°, and 60° respectively.

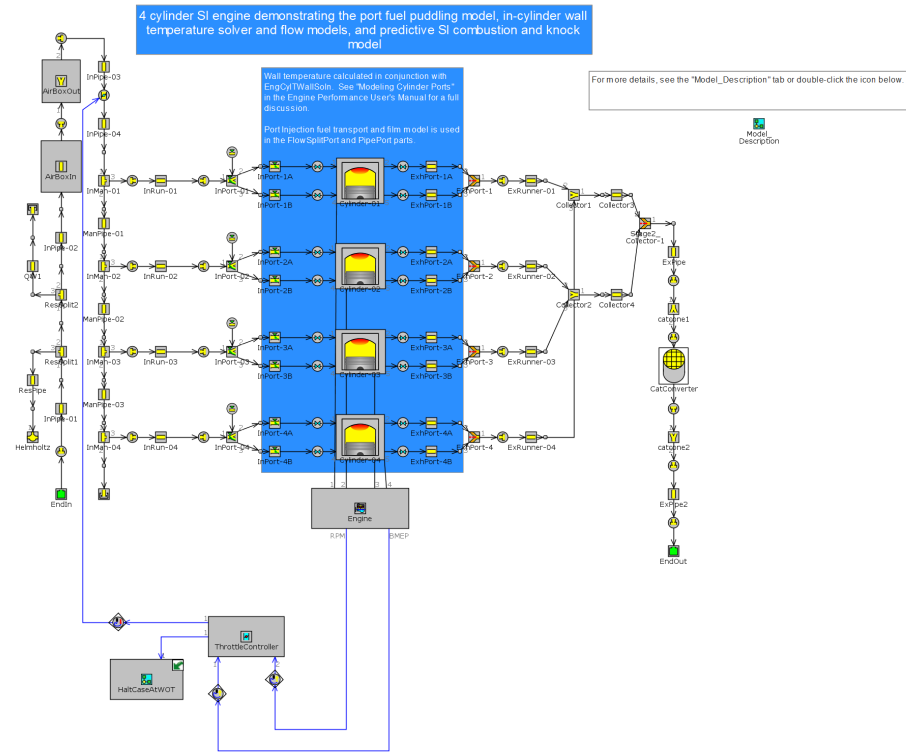


Figure 2: Case 4 model: the base engine model augmented with a closed-loop throttle controller (feedback on engine speed and BMEP, output = throttle angle) and a HaltCaseAtWOT safety block.

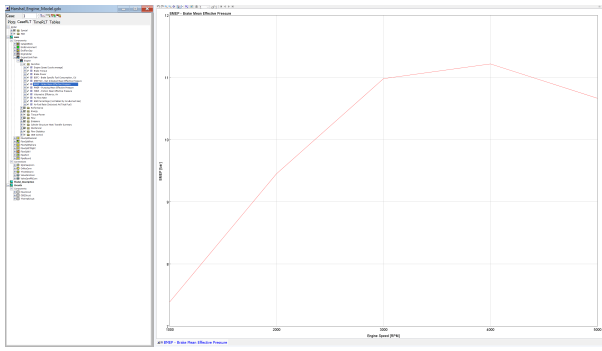
Throughout the report the standard mean-effective-pressure formalism developed in the course lecture notes is used to interpret the trends: indicated work $W_i = \oint p dV$, IMEP = W_i/V_d , BMEP = W_b/V_d , brake power $P_b = T_b \omega = \text{BMEP} \cdot V_d \cdot n / (60 \varepsilon_t)$, mechanical efficiency $\eta_{mech} = \text{BMEP}/\text{IMEP}$, BSFC = $\dot{m}_f/P_b$, and volumetric efficiency $\lambda_v = V_{actual}/V_{max}$ (formulas F8–F16 of the course compendium).

2 Case 1 – Engine Speed Sweep (Baseline, Stoichiometric AFR)

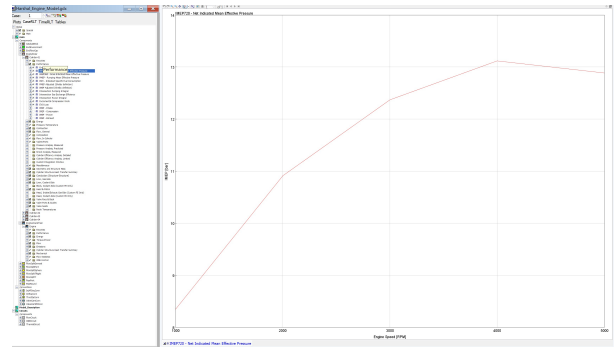
2.1 Results

Table 1: Case 1 – performance parameters vs. engine speed

RPM	1000	2000	3000	4000	5000
BMEP [bar]	7.40	9.55	11.00	11.25	10.60
IMEP _{net} [bar]	8.40	10.90	12.40	13.10	12.85
Brake torque [N·m]	116.5	148.5	172.5	176.0	166.5
Brake power [kW]	12.2	31.5	55.0	74.5	88.5
BSFC [g/kWh]	269	243.5	240	245.5	259.5
Ind. efficiency [%]	34.4	38.2	39.0	38.5	34.5
Vol. efficiency, cyl. 1–4 [-]	≈0.69	0.79–0.85	0.90–0.95	0.95–0.97	0.93–1.00

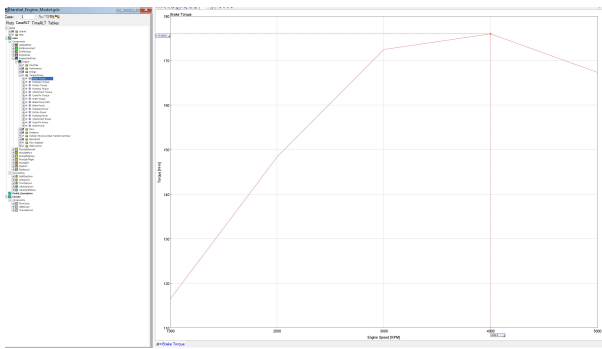


(a) BMEP vs. engine speed

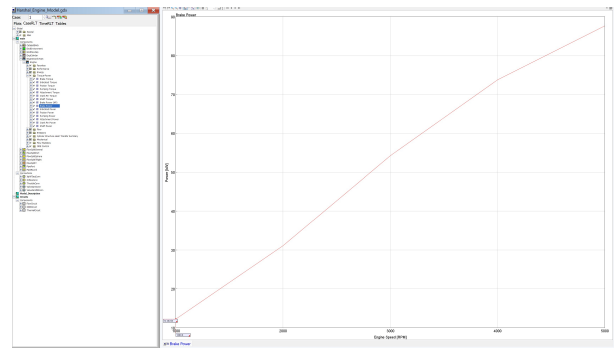


(b) Net IMEP vs. engine speed

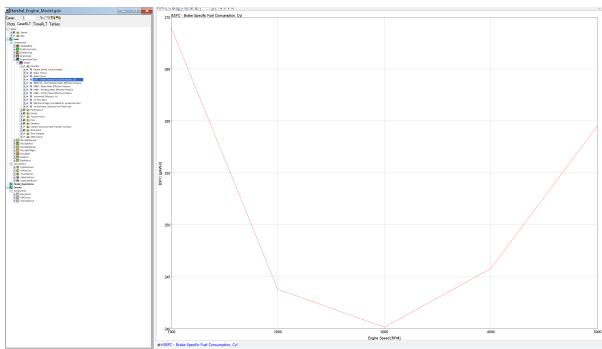
Figure 3: Case 1: mean effective pressures vs. engine speed.



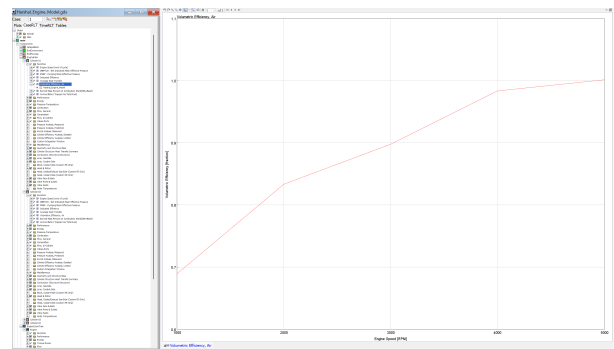
(a) Brake torque vs. engine speed



(b) Brake power vs. engine speed

Figure 4: Case 1: torque and power curves. Power keeps rising through 5000 rpm because $P_b = T_b \omega$ even after torque itself peaks at 4000 rpm.

(a) BSFC vs. engine speed



(b) Cylinder-1 volumetric efficiency vs. engine speed

Figure 5: Case 1: BSFC and volumetric efficiency vs. engine speed.

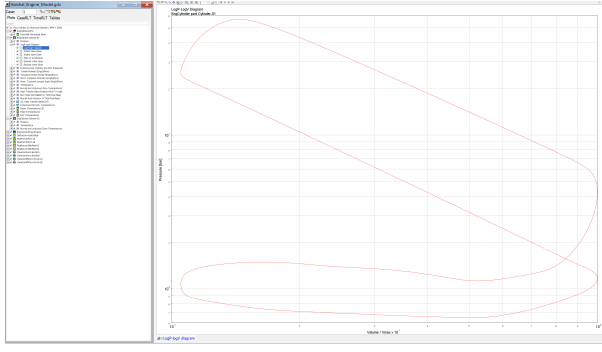
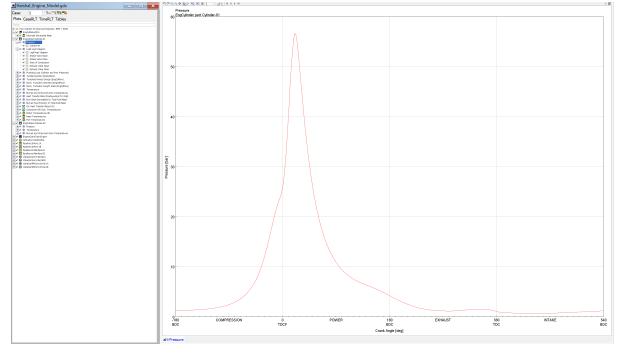
(a) p - V diagram (5000 rpm)(b) p - θ diagram (5000 rpm)

Figure 6: Case 1 indicator diagrams at 5000 rpm: classic figure-eight log p -log V loop (large high-pressure compression/combustion/expansion loop plus a small low-pressure gas-exchange loop), and peak firing pressure ≈ 57 bar a few degrees after TDC.

2.2 Discussion

Discussion: Torque and BMEP rise steeply from 1000 to 3000 rpm and peak around 3500–4000 rpm before falling off toward 5000 rpm, while brake power keeps rising monotonically through the whole range. This is exactly the shape predicted by combining $T_b = \text{BMEP} \cdot V_d / (2\pi\epsilon_t)$ with $P_b = T_b \omega$ (F1, F11, F12): at low speed the volumetric efficiency λ_v (F15) is depressed by low intake-manifold dynamic pressure and poor scavenging, so less air/fuel mass is trapped per cycle and BMEP is low. As speed increases, ram/tuning effects raise λ_v (visible directly in the cylinder-1 volumetric-efficiency curve, which rises almost monotonically from ≈ 0.69 at 1000 rpm to ≈ 1.0 by 5000 rpm), so more air is trapped and BMEP climbs. Beyond the torque peak (≈ 4000 rpm), frictional and pumping losses (FMEP, F13) grow roughly with the square of piston speed, eroding $\eta_{\text{mech}} = \text{BMEP} / \text{IMEP}$ faster than λ_v can keep improving, so BMEP – and hence torque – turns over even though IMEP itself is still close to its own peak. Brake power, being torque *times* the still-increasing $\omega = 2\pi n / 60$ (F11), continues to rise past the torque peak until the torque roll-off finally outpaces the rise in ω – this is precisely the mechanism described by F17 for why T_{max} always occurs at a lower speed than $P_{b,\text{max}}$.

BSFC (F14, $= \dot{m}_f / P_b = 1 / (\eta_{\text{engine}} \cdot \text{LHV})$) traces the mirror image of the indicated-efficiency curve: it is minimized (≈ 240 g/kWh) near 3000 rpm where indicated efficiency peaks ($\approx 39\%$), and rises at both ends of the speed range – at low speed because of relatively larger heat-transfer and friction losses per unit of (small) charge trapped, and at high speed because rising FMEP and non-ideal (shortened) combustion duration relative to crank-angle both reduce η_{engine} .

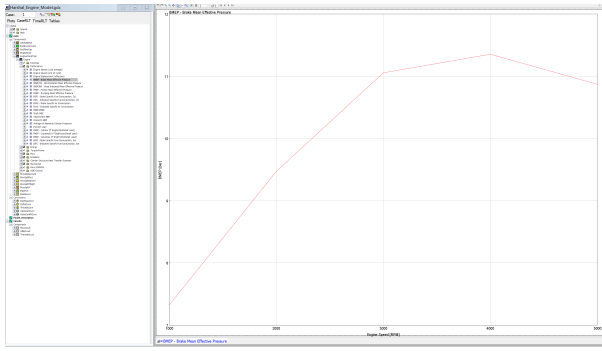
3 Case 2 – Air–Fuel Ratio Sweep (Rich vs. Lean)

3.1 Results

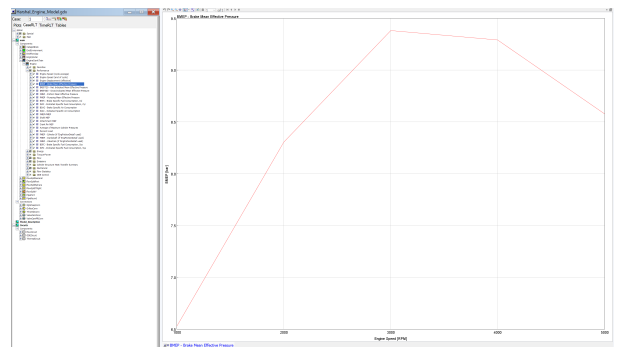
Table 2: Case 2 – BMEP [bar] and BSFC [g/kWh] vs. engine speed for rich (AFR 11.76) and lean (AFR 17.64) mixtures

RPM	1000	2000	3000	4000	5000
BMEP, rich [bar]	7.50	9.50	11.05	11.40	10.65
BMEP, lean [bar]	6.50	8.35	9.35	9.20	8.35
BSFC, rich [g/kWh]	342	304	297	303	316
BSFC, lean [g/kWh]	259	234	236	249	267
Brake torque, rich [N·m]	114	148	171	177	168
Brake torque, lean [N·m]	102	129	145	143	130
Brake power, rich [kW]	12	31	52	74	89
Brake power, lean [kW]	10.5	27.5	46.5	61	71

For reference, the Case-1 stoichiometric BMEP peak was ≈ 11.25 bar (4000 rpm) – almost exactly between the lean and rich extremes, as expected.

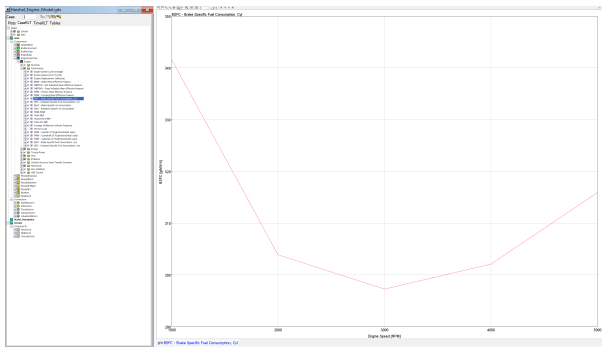


(a) BMEP, rich mixture (AFR 11.76)

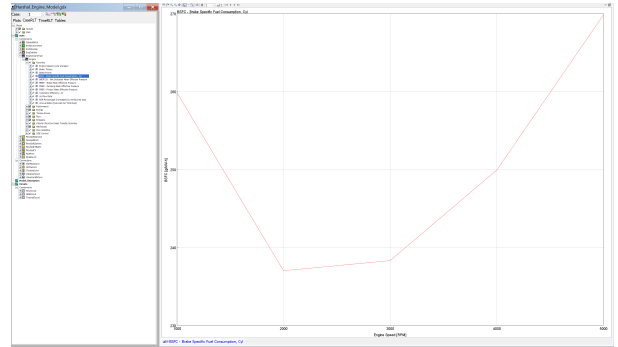


(b) BMEP, lean mixture (AFR 17.64)

Figure 7: Case 2: BMEP vs. engine speed, rich vs. lean mixture.



(a) BSFC, rich mixture



(b) BSFC, lean mixture

Figure 8: Case 2: BSFC vs. engine speed, rich vs. lean mixture.

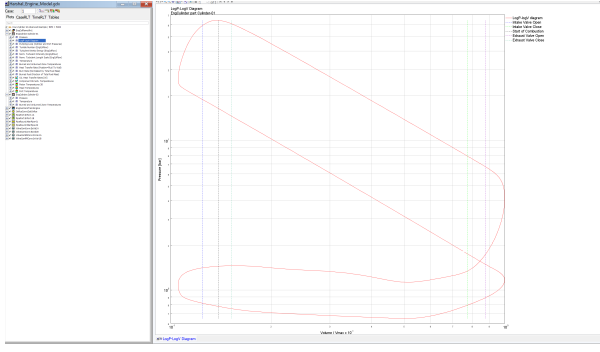
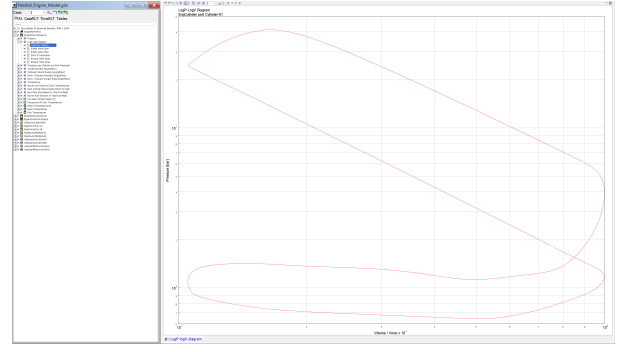
(a) p - V diagram, rich mixture(b) p - V diagram, lean mixture

Figure 9: Case 2 indicator diagrams: the rich-mixture loop encloses a visibly larger high-pressure area (more indicated work per cycle) than the lean-mixture loop.

3.2 Discussion

Combustion mixture ratio

$$\text{AFR} = \frac{\dot{m}_{\text{air}}}{\dot{m}_{\text{fuel}}}, \quad \lambda_{\text{comb}} = \frac{\text{AFR}}{\text{AFR}_{\text{stoich}}}, \quad \text{AFR}_{\text{stoich, gasoline}} \approx 14.7. \quad (1)$$

Discussion: With λ_v (breathing) and engine speed both fixed relative to Case 1, the trapped air mass $\dot{m}_{\text{air}} = \rho_{\text{amb}} V_d (n/60\varepsilon_t) \lambda_v$ (F16) is essentially unchanged between rich, stoichiometric and lean runs; what changes is $\dot{m}_{\text{fuel}} = \dot{m}_{\text{air}}/\text{AFR}$. Enriching the mixture (AFR= 11.76, $\lambda_{\text{comb}} \approx 0.80$) injects $\approx 25\%$ more fuel than stoichiometric for the same trapped air, releasing more chemical energy per cycle and raising IMEP/BMEP – this is the direct explanation for the rich curve sitting above both the Case-1 baseline and the lean curve at every speed in the table. Conversely, leaning the mixture (AFR= 17.64, $\lambda_{\text{comb}} \approx 1.20$) starves the same air charge of fuel, reducing the energy released per cycle and depressing BMEP, torque and power by 15–25% relative to stoichiometric.

BSFC moves in the opposite direction: because $\text{BSFC} = \dot{m}_f/P_b = 1/(\eta_{\text{engine}} \cdot \text{LHV})$ (F14) is governed by *efficiency* rather than by absolute fuel flow, and lean combustion in SI engines is thermodynamically more efficient near $\lambda_{\text{comb}} \approx 1.1$ – 1.2 (closer to the theoretical MBT/lean-limit efficiency optimum, less unburned excess fuel, lower pumping losses at a given load), the lean case shows BSFC ≈ 20 – 25% *lower* than the rich case across the whole speed range, even though its absolute power output is lower. This is the classic rich-power vs. lean-economy trade-off: rich mixtures buy peak torque/power at the cost of fuel economy, lean mixtures buy fuel economy at the cost of peak output – both are direct, opposite-signed consequences of the same $\dot{m}_f = \dot{m}_{\text{air}}/\text{AFR}$ relationship acting on the numerator (fuel consumed) vs. the denominator (power produced) of the BSFC definition. Volumetric efficiency itself is essentially unaffected by AFR (both curves track the Case-1 breathing curve to within ± 0.02), confirming that λ_v is governed by intake/valve geometry and speed, not by the fuelling ratio, exactly as F15 implies.

4 Case 3 – Spark Advance Sweep (Fixed 3000 rpm)

4.1 Results

Table 3: Case 3 – peak in-cylinder pressure read from the p - θ diagrams at 3000 rpm, all five spark-advance settings

Spark advance setting	-10°CA	-15°CA	-20°CA	-25°CA	-30°CA
Peak cylinder pressure	≈ 40 bar	≈ 51 bar	≈ 61 bar	≈ 69 bar	≈ 76 bar
Crank angle of peak pressure	$\approx 29^\circ$ ATDC	$\approx 18^\circ$ ATDC	$\approx 10^\circ$ ATDC	$\approx 7^\circ$ ATDC	$\approx 7.8^\circ$ ATDC

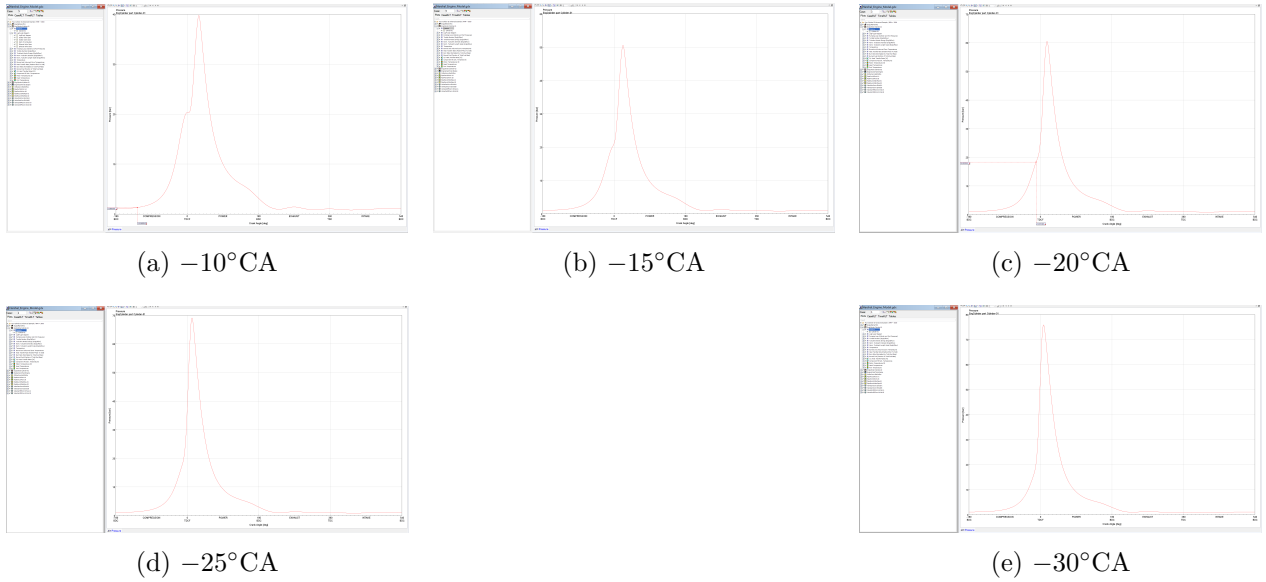


Figure 10: Case 3: pressure vs. crank angle at 3000 rpm, all five spark-advance settings. Advancing the spark from -10°CA to -30°CA moves combustion progressively earlier relative to TDC, monotonically raising peak pressure ($40 \rightarrow 76$ bar) while moving its location steadily closer to TDC ($29^{\circ} \rightarrow 7-8^{\circ}$ ATDC).

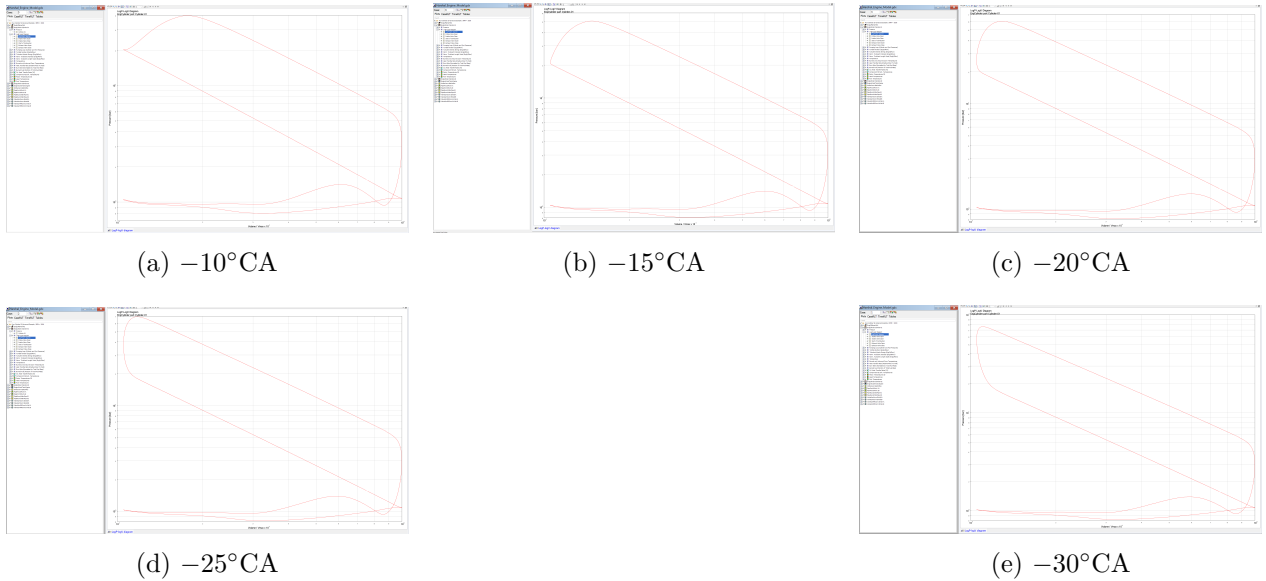


Figure 11: Case 3: $\log p$ - $\log V$ diagrams at all five spark-advance settings; the enclosed high-pressure loop area grows steadily from -10°CA to -30°CA , consistent with the monotonically increasing indicated work implied by the rising peak pressures above.

4.2 Discussion

Discussion: In the sign convention used by this GT-Power model, a spark-advance value such as -30°CA means ignition is triggered 30° of crank angle *before* TDC – i.e. a larger magnitude corresponds to a more *advanced* spark, while -10°CA (smaller magnitude) corresponds to a more *retarded* spark relative to the sweep’s other settings. Because the flame-development and flame-propagation process inside the cylinder always takes a roughly fixed amount of crank-angle time at a given speed, firing earlier (more advanced, -30°CA) gives the flame more time to complete combustion close to TDC, producing a sharp pressure rise concentrated near minimum cylinder volume – exactly the near-TDC, high-magnitude peak (≈ 76 bar at $\approx 8^{\circ}$ ATDC) seen in the -30°CA trace. Retarding the spark toward -10°CA delays ignition, so combustion occurs increasingly during the expansion stroke where cylinder volume is already growing; per IMEP =

$\oint p dV/V_d$ (F8, F9), releasing the same chemical energy over a larger, later volume produces a lower, later, more spread-out pressure peak (≈ 40 bar at $\approx 29^\circ$ ATDC), and correspondingly lower enclosed p - V loop area, i.e. lower IMEP/BMEP and lower torque – clearly visible in the smaller loop area of the -10° CA log p -log V diagram compared with the -30° CA case.

This is the standard spark-timing/MBT (Minimum-advance-for-Best-Torque) trade-off: firing too late (retarded) wastes combustion energy on the expansion stroke instead of building pressure while the piston is near TDC, lowering IMEP, BMEP, torque and raising BSFC; firing progressively earlier recovers torque up to an MBT optimum, beyond which further advance pushes peak pressure too early in the compression stroke, increasing negative compression work and knock risk and eventually reducing net torque again. The pressure and timing data extracted here place the -30° CA setting closer to this MBT/near-optimal region than -10° CA, consistent with -30° CA producing a substantially larger indicated-work loop.

5 Case 4 – Throttle Position / Load Sweep

5.1 Results

Table 4: Case 4 – BSFC vs. throttle angle / load (from simulation output data)

Load	5%	10%	20%	50%
Throttle angle $[\circ]$	18	26	37	60
BSFC [g/kWh]	1126.0	653.2	421.2	282.5
Peak cylinder pressure [bar] (from p - θ)	≈ 8.4	≈ 10.6	≈ 15.0	≈ 29.5

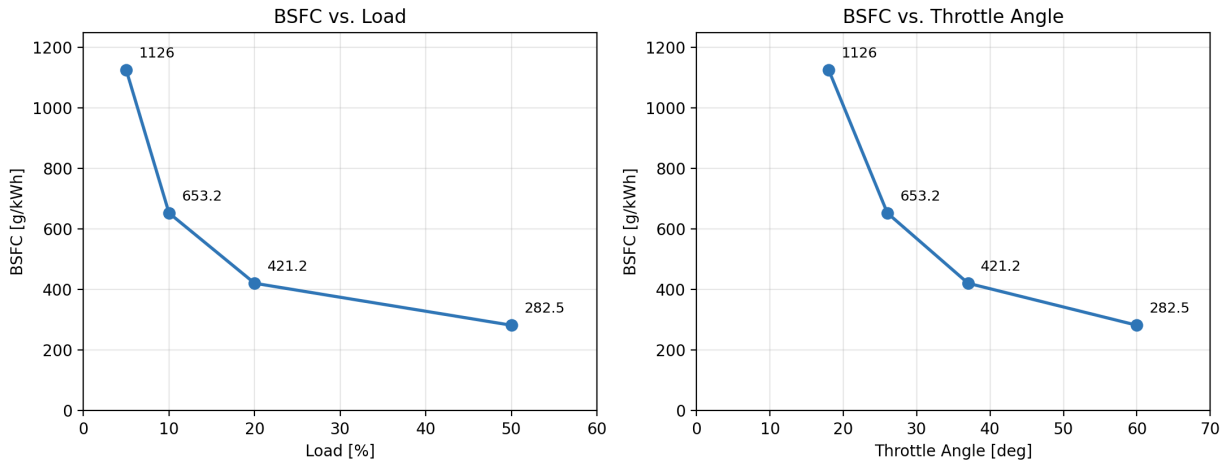


Figure 12: Case 4: BSFC as a function of load (%) and of throttle angle ($^\circ$), plotted from the same four data points as the table above. Both curves have an identical shape (load and throttle angle are in one-to-one correspondence for this sweep) – a steep, strongly non-linear drop in BSFC from 1126 g/kWh at 5% load down to 282.5 g/kWh at 50% load, with more than half of the total improvement already realised by the second point (10% load, 653.2 g/kWh).

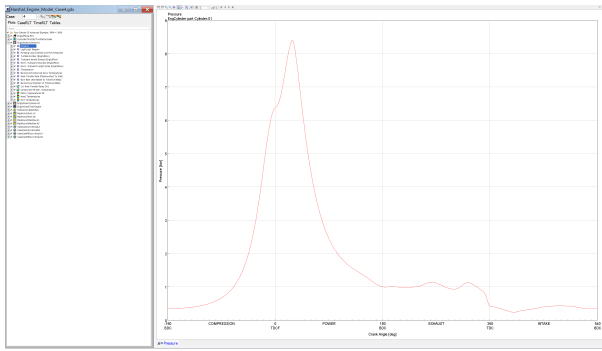
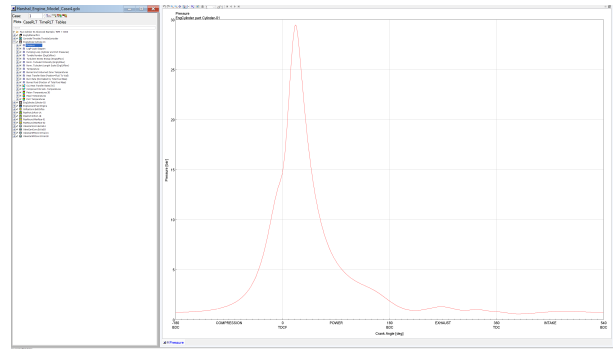
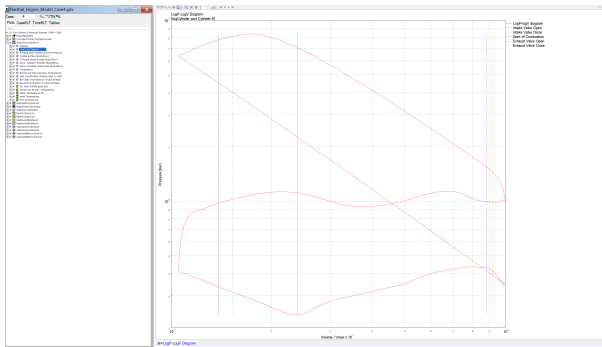
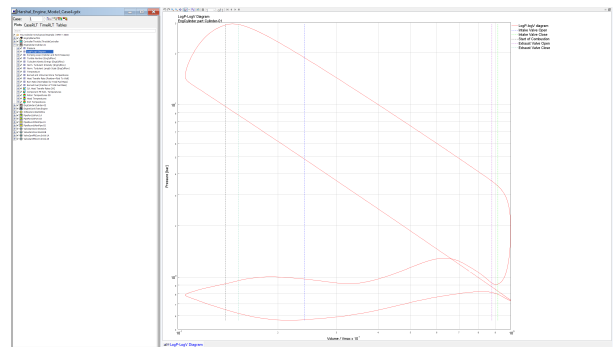
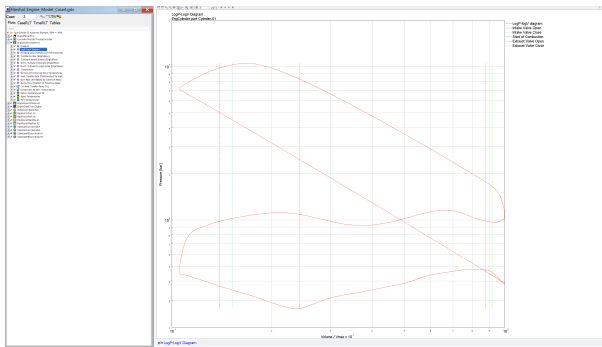
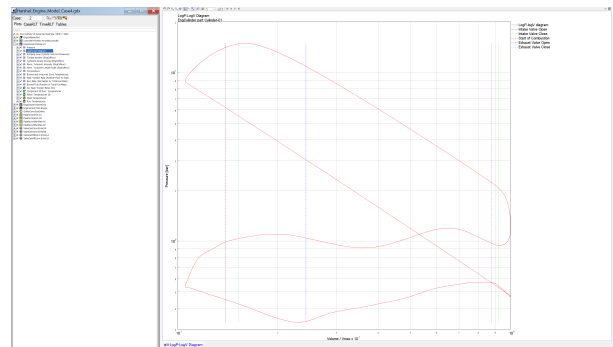
(a) $p-\theta$, 5% load(b) $p-\theta$, 50% load

Figure 13: Case 4: pressure vs. crank angle at the lowest (5%) and highest (50%) load points tested.

(a) $p-V$, 5% load(b) $p-V$, 50% loadFigure 14: Case 4 indicator diagrams: at 5% load the low-pressure gas-exchange (pumping) loop is large and deep (intake pressure well below atmospheric, throttle plate nearly closed at 18°); at 50% load the pumping loop has shrunk to a thin, near-atmospheric band while the high-pressure loop has grown dramatically ($18^\circ \rightarrow 60^\circ$ throttle opening).(a) $p-V$, 10% load(b) $p-V$, 20% loadFigure 15: Intermediate load points (10%, 20%): the pumping loop shrinks and the high-pressure loop grows progressively as the throttle opens from 26° to 37° .

5.2 Discussion

Throttling / pumping loss

$$\text{FMEP} = \text{IMEP} - \text{BMEP} = \text{PMEP} + \text{FMEP}_{\text{mech}}, \quad \text{PMEP} \propto (p_{\text{atm}} - p_{\text{intake, manifold}}). \quad (2)$$

Discussion: At light load the throttle plate is nearly closed (18° at 5% load), so the engine must pull air through a highly restricted opening: intake manifold pressure drops well below atmospheric, and the piston does substantial negative work dragging air in against this vacuum during the intake stroke – this is exactly the large, deep low-pressure loop seen in the 5% p - V diagram, and it directly enlarges the pumping mean effective pressure (PMEP) component of FMEP (F13). As the throttle opens toward wide-open throttle (WOT, 60° at 50% load), the pressure drop across the throttle plate collapses, intake pressure approaches atmospheric, and the pumping loop shrinks to a thin near-atmospheric band – reducing PMEP and improving $\eta_{\text{mech}} = \text{BMEP}/\text{IMEP}$.

Simultaneously, opening the throttle raises the trapped air mass and hence peak cylinder pressure once combustion occurs (peak pressure rises monotonically and steeply from ≈ 8.4 bar at 5% load to ≈ 29.5 bar at 50% load), directly increasing indicated work per the $\oint p dV$ definition (F8) and therefore IMEP and brake power at fixed speed. BSFC falls sharply and monotonically with load (1126 g/kWh at 5% down to 282.5 g/kWh at 50%) because $\text{BSFC} = \dot{m}_f / P_b$ (F14): fuel flow scales roughly with air flow (fixed AFR), which itself scales only mildly with throttle opening at fixed speed, while P_b grows much faster because it benefits from *both* the reduced pumping losses (higher η_{mech}) *and* the higher IMEP – so the denominator of BSFC grows much faster than the numerator. This is the standard part-load throttling-loss penalty familiar from vehicle drive-cycle analysis: naturally-aspirated SI engines are intrinsically least efficient at low load/high manifold vacuum, and become progressively more efficient as the throttle opens toward WOT, exactly as reproduced here.

Discussion: The shape of the BSFC curve in Fig. 12 is itself diagnostic. The BSFC-vs-load relationship is strongly *convex/non-linear* rather than a straight line: going from 5% to 10% load (an increase of only 5 percentage points, or 8° of throttle opening) already cuts BSFC almost in half ($1126 \rightarrow 653.2$ g/kWh, a 42% reduction), whereas the next 30-percentage-point increase from 20% to 50% load only improves BSFC by a further 33% ($421.2 \rightarrow 282.5$ g/kWh). This asymmetry follows directly from the PMEP mechanism above: the pumping-loop area (and hence PMEP) is governed by the pressure ratio across the nearly-closed throttle plate, which collapses very quickly as the plate first starts to open (small angles) and much more gradually thereafter as the flow area becomes progressively less restrictive – i.e. the relationship between throttle angle and effective flow area is itself non-linear near closure. The practical implication, directly relevant to the hybrid-vehicle energy-management strategy studied in Exercise 2, is that the *first* increment of load taken away from idle/light-load operation buys by far the largest efficiency gain – exactly the reasoning that motivates an HEV’s ECMS supervisory controller to avoid running the ICE at very light load at all, preferring to keep it either off (electric-only propulsion) or loaded into the flatter, already-efficient region of this same curve.

6 Overall Conclusions

Across all four sub-cases, every observed trend follows directly from the same small set of governing relationships established in the course notes: brake torque and BMEP are set by how much air/fuel mass is trapped and burned per cycle relative to displacement (λ_v , AFR, spark timing all act on this), brake power is torque multiplied by the ever-increasing angular speed ω (so power keeps climbing after torque itself has peaked), and BSFC is the inverse of overall thermal efficiency and therefore captures every source of loss – pumping (throttling), mechanical friction, and combustion-phasing losses – in a single measurable number.

- **Case 1 (speed sweep):** torque/BMEP rise with improving volumetric efficiency, peak near 3500–4000 rpm once rising friction/pumping losses overtake the breathing gains, while

power keeps rising until ≈ 5000 rpm; BSFC is minimized where indicated efficiency peaks, near 3000 rpm.

- **Case 2 (AFR sweep):** enrichment trades fuel economy for peak torque/power (more fuel energy released per cycle for the same trapped air); leaning trades peak output for fuel economy – a direct, opposite-signed consequence of $\dot{m}_f = \dot{m}_{air}/\text{AFR}$ acting on BSFC's numerator vs. P_b .
- **Case 3 (spark advance sweep):** advancing the spark concentrates combustion closer to TDC, raising peak pressure and enclosed indicated-work area (higher torque) up to an MBT-like optimum; retarding it spreads combustion into the expansion stroke, lowering peak pressure and indicated work.
- **Case 4 (throttle/load sweep):** closing the throttle at part load creates a large pumping loop (negative work, high P_{MEP}) that both lowers mechanical efficiency and starves the cylinder of trapped air (low peak pressure, low IMEP), compounding to give very high BSFC at light load; opening the throttle toward WOT collapses the pumping loop and sharply improves BSFC.

All four sets of results are therefore internally consistent with, and directly explained by, the mean-effective-pressure and power-chain formalism (F1, F8–F17) developed in the course.

Vehicle Simulation of a P1 Hybrid Electric Vehicle with ECMS Energy Management

Simulation over WLTC Class 3 and Artemis driving cycles,
emission assessment and vehicle-mass sensitivity analysis

Harshal Rajesh Kokitkar

Model No.: P1 — ICE maximum torque multiplier: 1.02

July 14, 2026

Abstract

This report presents the simulation of a P1 parallel hybrid electric vehicle (HEV) along the WLTC Class 3 driving cycle, controlled by an Equivalent Consumption Minimization Strategy (ECMS). The vehicle model is based on the MathWorks Powertrain Blockset HEV P1 reference application (Virtual Vehicle Composer), with the maximum torque of the internal combustion engine scaled by the individually assigned multiplier of 1.02. The constant equivalence factor was tuned to $s = 3.23$ to achieve battery state-of-charge (SOC) balance over the WLTC, and the simulations were then performed with the adaptive ECMS formulation. Without modifying the ECMS settings, the vehicle was further simulated on the Artemis Rural Road and Artemis Motorway 130 km/h cycles, and the tailpipe emissions were compared against the gasoline Euro 6e limits and the EU fleet CO₂ target for 2025. The instantaneous operating points of the engine and of the electric machine were extracted and analysed over the engine BSFC map and the motor efficiency map, respectively. Finally, a sensitivity analysis on the vehicle mass ($\pm 25\%$) was carried out and the results are justified on a theoretical basis.

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1 Introduction

Hybrid electric vehicles combine an internal combustion engine (ICE) with one or more electric machines (EM) and an energy storage system, with the goal of reducing fuel consumption and pollutant emissions. In the *P1* architecture the electric machine is rigidly connected to the engine crankshaft, downstream of the engine and upstream of the clutch and transmission. Consequently:

- the EM always rotates with the engine (no pure electric drive is possible while the vehicle moves with the ICE decoupled);
- the EM can perform torque assist, generation (load-point shifting) and partial regenerative braking through the driveline;
- start/stop operation is efficient because the EM doubles as an integrated starter-generator.

Because the P1 machine cannot propel the vehicle alone, the energy management strategy acts essentially on the *torque split* between ICE and EM at the crankshaft, deciding instant by instant whether the electric machine should assist (battery discharge) or generate (battery charge), and how much.

The simulations described in this report were carried out with the MathWorks Virtual Vehicle Composer / Powertrain Blockset *HEV P1 reference application* (`ConfiguredHevP1VirtualVehicle`), which contains:

- a **Scenario** block providing the drive cycle (WLTC Class 3, Artemis Rural Road, Artemis Motorway 130 km/h);
- a **Driver** model (longitudinal PI driver) converting the velocity trace into accelerator and brake commands;
- a **Controllers** subsystem containing the supervisory hybrid controller with the *Equivalent Consumption Minimization Strategy* (ECMS) block;
- a **Vehicle** plant with mapped SI engine, electric machine, battery, transmission and longitudinal vehicle dynamics, including tailpipe emission maps (HC, CO, NO_x, CO₂).

According to the individual assignment (Model No. P1), the maximum torque map of the ICE was scaled by a multiplier of **1.02** (i.e. +2% on the brake torque map).

2 Energy Management: ECMS

2.1 Working principle

The ECMS is an instantaneous optimization strategy derived from Pontryagin's Minimum Principle. At every controller time step it selects the torque split that minimizes an *equivalent* fuel mass flow rate, i.e. the sum of the actual fuel consumption of the engine and of a fuel equivalent of the electrical power exchanged with the battery:

$$\dot{m}_{f,eq}(t) = \dot{m}_f(t) + \frac{s(t)}{H_{LHV}} P_{batt}(t) f(\text{SOC}) \quad (1)$$

where $\dot{m}_f$ is the engine fuel mass flow rate, P_{batt} the battery electrochemical power (positive when discharging), H_{LHV} the fuel lower heating value and $s(t)$ the **equivalence factor** (EF). The penalty function $f(\text{SOC})$ increases the cost of electrical energy as the SOC departs from its target, keeping the battery within its admissible window. Because the P1 hybrid is charge

sustaining (no external recharge), all the energy ultimately comes from the fuel; the EF represents the ratio between the fuel that will be needed in the future to restore the battery charge and the electrical energy used now.

The role of the EF is central:

- **high** s : electrical energy is expensive $\rightarrow$ the strategy uses the ICE more, the battery charges, and the final SOC rises;
- **low** s : electrical energy is cheap $\rightarrow$ the EM assists more, and the battery depletes.

2.2 Tuning of the constant EF and adaptive ECMS

In the Powertrain Blockset ECMS block the constant EF corresponds to the *ECMS weighting factor* ECMS_s (Energy Management tab). The factor was tuned iteratively over the WLTC Class 3 cycle until the SOC balance condition

$$\Delta\text{SOC} = \text{SOC}(t_f) - \text{SOC}(t_0) \approx 0 \quad (2)$$

was satisfied, obtaining

$$\boxed{s = 3.23} \quad (3)$$

The simulations reported here were then executed with the *adaptive* ECMS option of the block, in which a PI controller corrects the equivalence factor as a function of the SOC error with respect to the target ($\text{SOC}_{\text{trg}} = 60\%$):

$$s(t) = s_0 + K_p (\text{SOC}_{\text{trg}} - \text{SOC}(t)) + K_i \int_0^t (\text{SOC}_{\text{trg}} - \text{SOC}(\tau)) d\tau \quad (4)$$

The adaptive formulation makes the strategy robust to changes of driving cycle and vehicle parameters: this is essential for the following sections, where the cycle and the vehicle mass are changed *without retuning the ECMS*.

3 Baseline results: WLTC Class 3

3.1 Cycle following, SOC and fuel economy

Figure 1 shows the velocity trace (target vs. actual), the battery current, the engine and motor speeds and torques, the battery SOC and the cumulative fuel economy over the WLTC Class 3 cycle (1800 s, 23.27 km). The driver model follows the target velocity closely over all four phases (low, medium, high, extra-high).



Figure 1: WLTC Class 3, baseline vehicle (1623 kg): velocity trace, battery current, engine/motor speed and torque, battery SOC and cumulative fuel economy.

Key observations:

- The SOC starts at 60%, oscillates between about 58% and 63% during the first three phases (charge-sustaining behaviour) and dips to about 55% during the extra-high-speed phase, where the power demand is highest; the adaptive EF then drives it back towards the target, ending at $\approx 57.5\text{--}58.5\%$. The residual ΔSOC is small compared to the battery energy content, so the fuel consumption figure is representative.
- The battery current shows frequent positive (discharge/assist) spikes during accelerations and negative (charge/regeneration) spikes during decelerations.
- The cumulative fuel economy converges to approximately 5.5 l/100 km.

3.2 ICE and EM power

Figure 2 reports the instantaneous ICE and EM mechanical power. The engine provides essentially all the traction power (peaks up to $\approx 50\text{ kW}$ in the extra-high phase), while the P1 electric machine exchanges comparatively small power levels (roughly $\pm 3\text{ kW}$): short positive bursts of torque assist during transients and negative power during regenerative phases and load-point shifting. This is the expected behaviour of a P1 mild-hybrid architecture, in which the degree of hybridization is small and the EM primarily smooths the engine operation and recovers braking energy through the driveline.

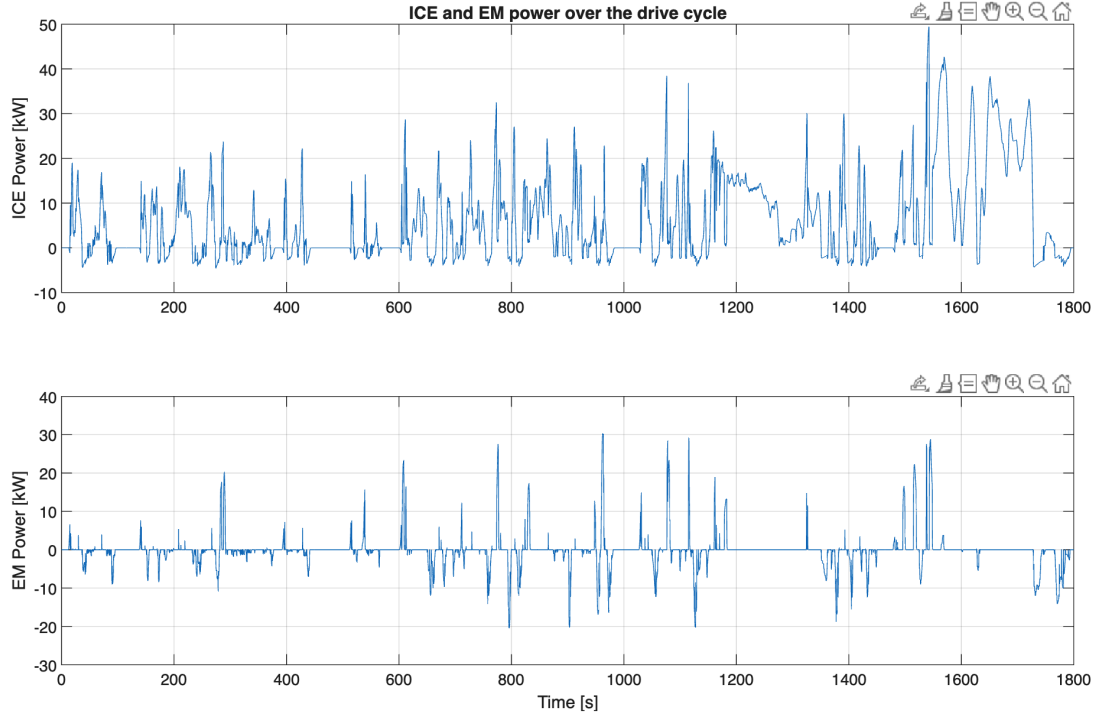


Figure 2: WLTC Class 3, baseline: instantaneous ICE power (top) and EM power (bottom).

3.3 Tailpipe emissions and comparison with Euro 6e and the EU CO₂ target

Figure 3 shows the cumulative distance-specific tailpipe emissions of HC, CO, NO_x and CO₂. The end-of-cycle values, read at $t = 1800$ s, are collected in Table 1.

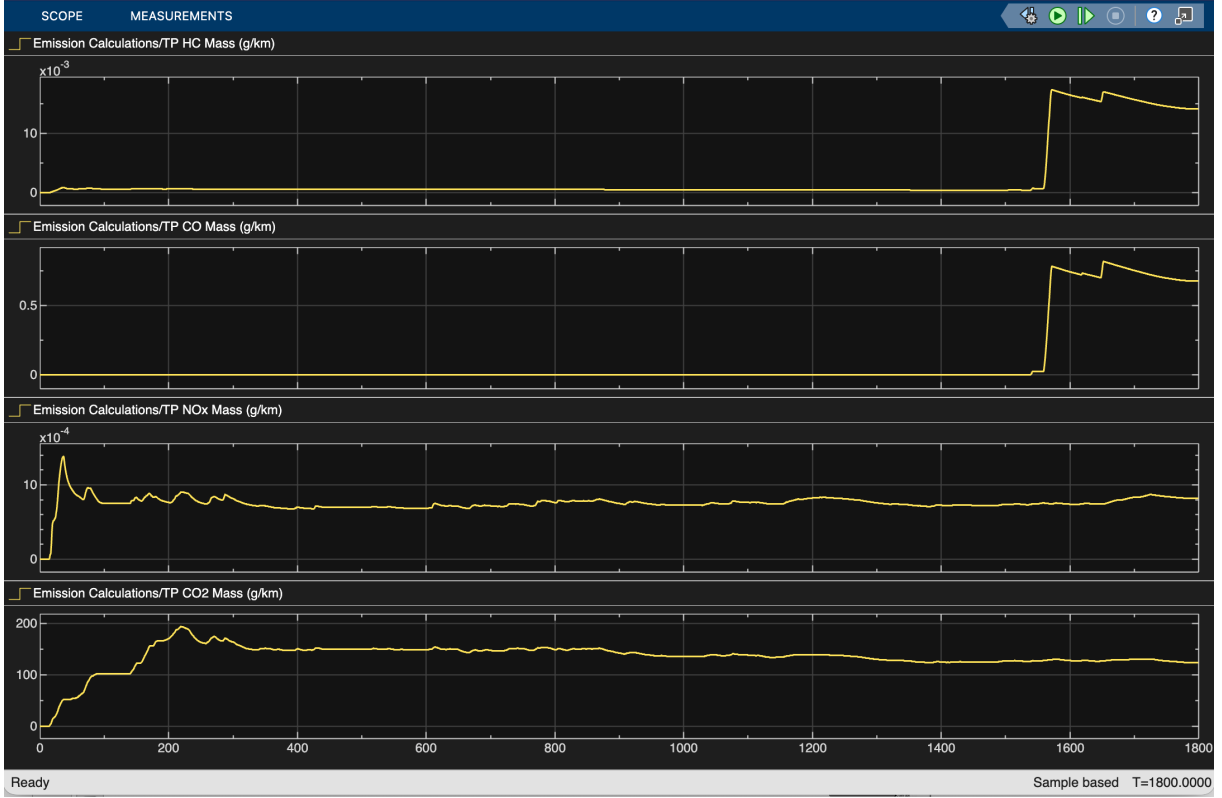


Figure 3: WLTC Class 3, baseline: cumulative tailpipe emissions of HC, CO, NO_x (g/km) and CO₂ (g/km).

Table 1: Tailpipe emissions over WLTC Class 3 (baseline vehicle) vs. gasoline Euro 6e limits and EU CO₂ fleet target. Values read from the emission scopes at end of cycle.

Pollutant	Simulated (WLTC)	Limit / target	Assessment
CO [g/km]	≈ 0.70	1.0 (Euro 6e)	compliant
THC [g/km]	≈ 0.014	0.10 (Euro 6e)	compliant
NO _x [g/km]	$\approx 8 \times 10^{-4}$	0.06 (Euro 6e)	compliant
CO ₂ [g/km]	≈ 125	93.6 (EU fleet target 2025, WLTP)	above target

Comments:

- HC, CO and NO_x are far below the Euro 6e limits. The catalyst operates at high conversion efficiency for almost the entire cycle; the visible step of HC and CO at $t \approx 1550$ s is caused by the high-load operation (fuel enrichment) of the extra-high-speed phase, not by a cold start.
- The CO₂ emission (≈ 125 g/km) exceeds the 2025 EU fleet average target of 93.6 g/km. It must be recalled that the EU target is a *fleet average* (manufacturer level), not a per-vehicle limit; nevertheless the result indicates that a mild P1 hybridization alone is not sufficient to reach the 2025 CO₂ levels for a vehicle of this mass, and stronger electrification (P2/P4 full hybrid, plug-in) or engine downsizing would be required.
- The small residual Δ SOC slightly flatters/penalizes the CO₂ figure; with SOC-corrected accounting the value would change by only a few g/km.

3.4 ICE operating points on the BSFC map

The instantaneous engine operating points (ω_{ICE}, T_{ICE}) logged during the cycle were superimposed on the Brake Specific Fuel Consumption (BSFC) map computed from the engine fuel map:

$$\text{BSFC} = \frac{\dot{m}_f}{P_b} = \frac{\dot{m}_f}{T_{ICE} \omega_{ICE}} \quad [\text{g/kWh}] \quad (5)$$

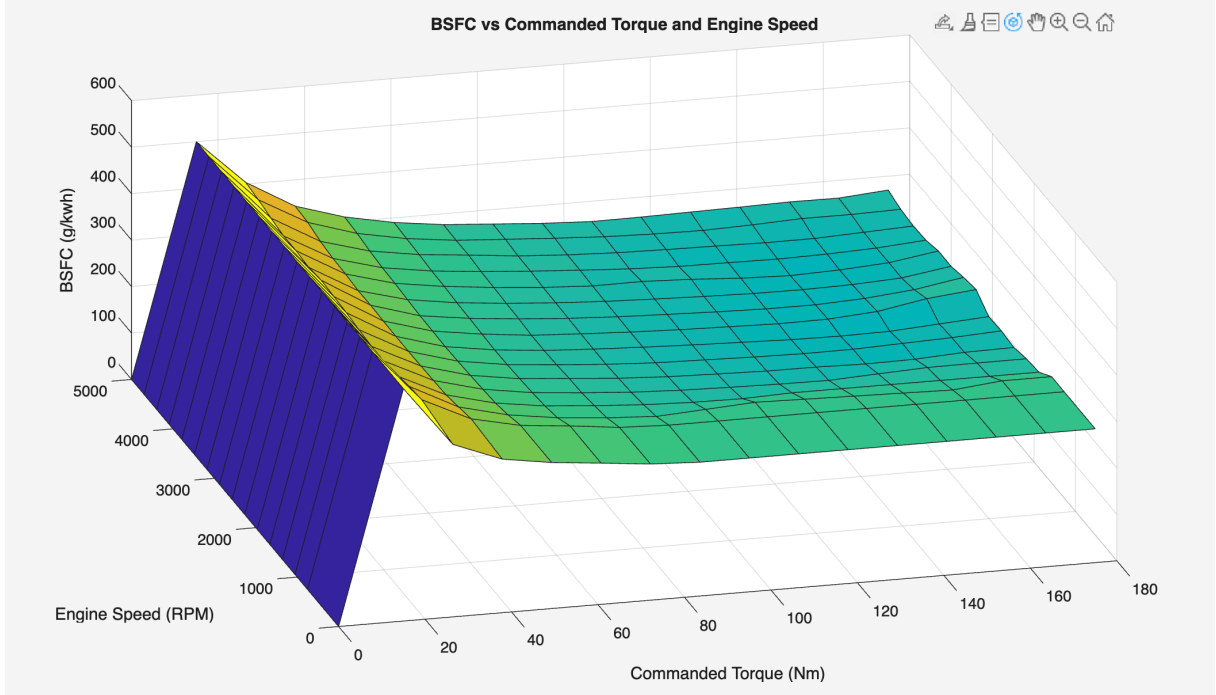


Figure 4: WLTC Class 3, baseline: instantaneous ICE operating points (red) over the BSFC map (g/kWh).

The point cloud (Figure 4) shows the load-shifting action of the ECMS:

- the engine operates mainly between 1000 and 3000 rpm, following the gear schedule of the transmission;
- a significant fraction of points is pushed towards medium-high loads, where the BSFC approaches its minimum ($\approx 240\text{--}260$ g/kWh), while the strategy avoids steady operation in the light-load region where the BSFC rises above 400 g/kWh;
- points along the full-load line (up to ≈ 175 Nm, consistent with the 1.02 torque multiplier) correspond to the strongest accelerations of the high and extra-high phases;
- points at nearly zero torque correspond to fuel cut and idle phases, where BSFC is not defined.

3.5 EM operating points on the efficiency map

Figure 5 shows the electric machine operating points over its efficiency map (mirrored to negative torque for generator operation).

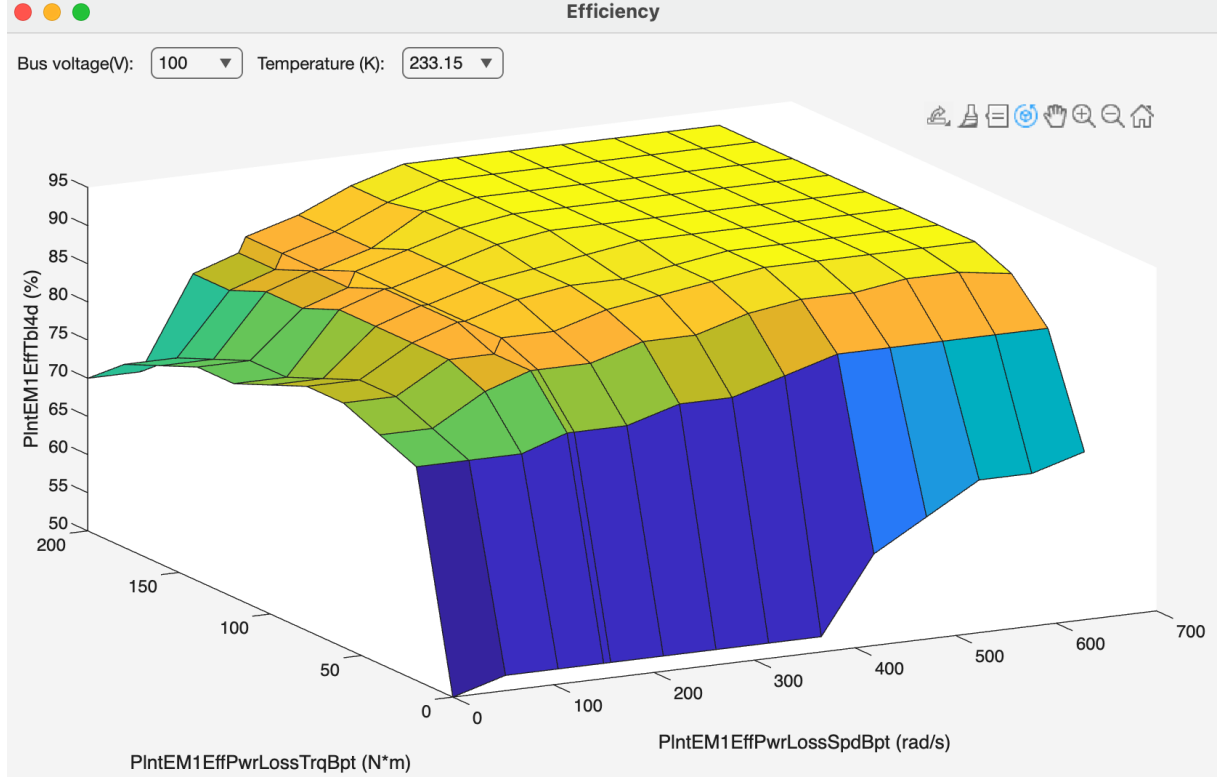


Figure 5: WLTC Class 3, baseline: EM operating points (white) over the motor efficiency map.

The EM points lie in a low-torque band on both sides of the zero-torque axis (motor assist above, generation/regeneration below), at speeds locked to the engine speed (P1 architecture). Because the exchanged torque is small, many points fall in the low-efficiency region of the map near zero torque; this is an intrinsic limitation of the P1 topology, in which the machine is sized for smoothing and recuperation rather than for traction, and it explains why the overall benefit of the electric path is modest compared to full-hybrid architectures.

4 Different driving cycles (same ECMS settings)

Without modifying the ECMS calibration, the vehicle was simulated on two additional cycles: **Artemis Rural Road** (1082 s, average speed ≈ 57 km/h) and **Artemis Motorway 130 km/h** (1068 s, cruise speeds up to 130 km/h). These two cycles bracket the WLTC in terms of load level.

4.1 Artemis Rural Road



Figure 6: Artemis Rural Road: velocity trace, battery current, speeds and torques, SOC, fuel economy.

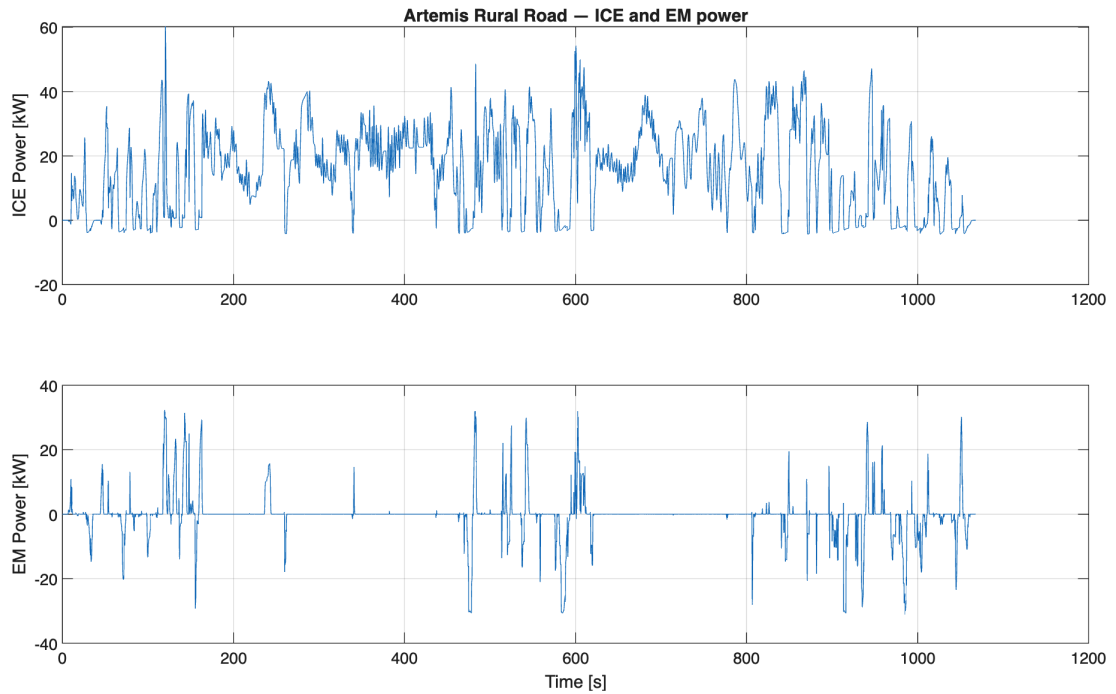


Figure 7: Artemis Rural Road: ICE and EM power vs time.

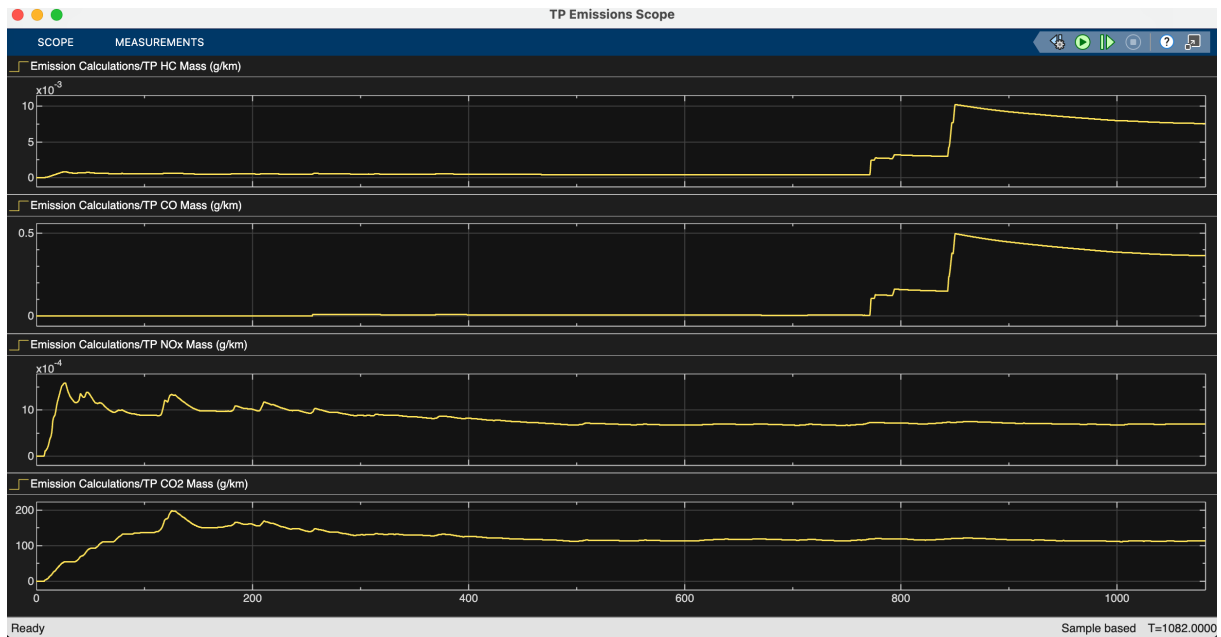


Figure 8: Artemis Rural Road: cumulative tailpipe emissions.

The rural cycle features frequent moderate accelerations and decelerations at medium speed: the SOC oscillates between 53% and 61%, ending near $\approx 57.7\%$, and the fuel economy settles around 5.01/100 km with $\text{CO}_2 \approx 115 \text{ g/km}$ — the best of the three cycles, because the operating points sit close to the efficient mid-load region and regeneration recovers a good share of the braking energy.

4.2 Artemis Motorway 130 km/h

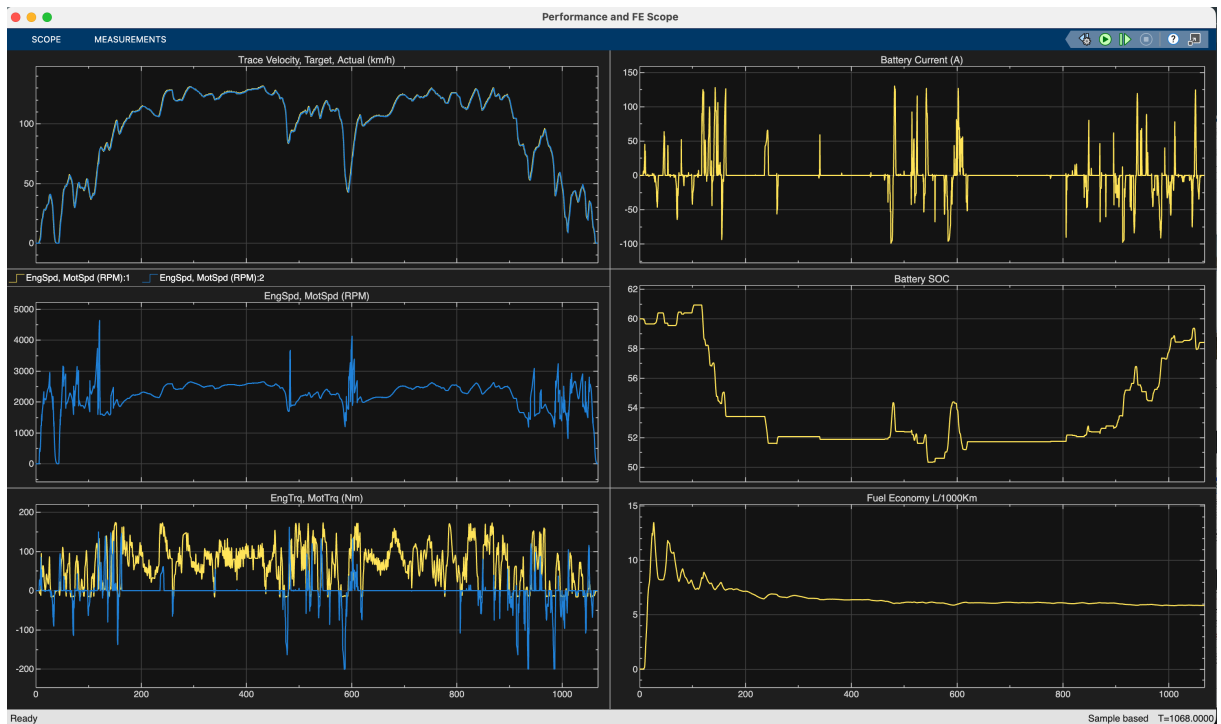


Figure 9: Artemis Motorway 130: velocity trace, battery current, speeds and torques, SOC, fuel economy.

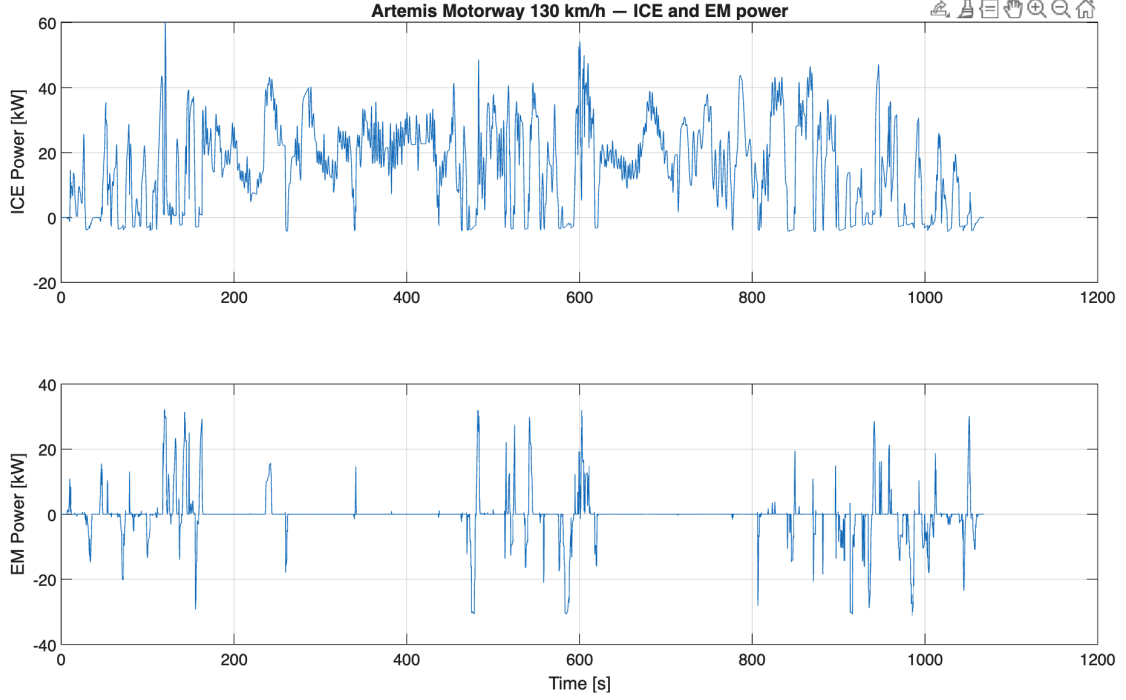


Figure 10: Artemis Motorway 130: ICE and EM power vs time.

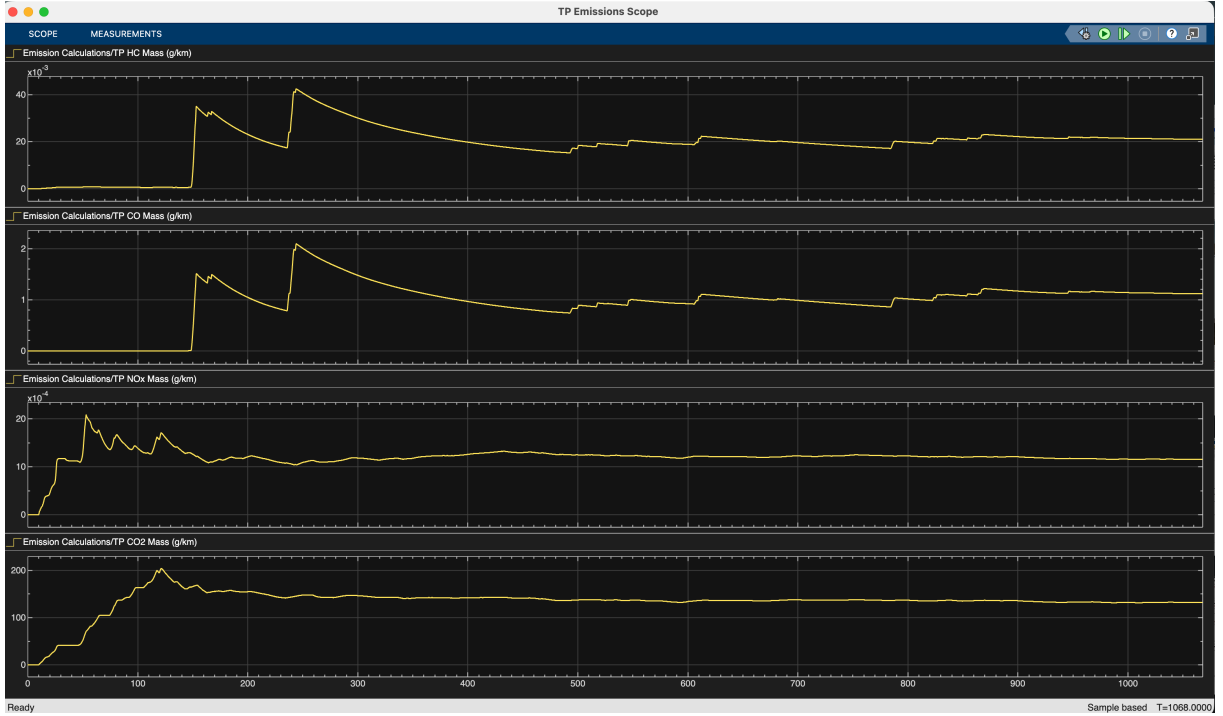


Figure 11: Artemis Motorway 130: cumulative tailpipe emissions.

On the motorway cycle the engine runs almost continuously at sustained load (60–130 Nm, 2000–2700 rpm) to overcome the aerodynamic resistance, which grows with the square of speed. The hybrid functionality is marginal: few engine-off phases, little regeneration. The SOC drops to $\approx 50.5\%$ during the fastest section and is restored by the adaptive EF towards $\approx 58.5\%$ at the end. Fuel economy worsens to $\approx 6.01/100$ km and CO_2 rises to ≈ 135 g/km. The HC and CO cumulative curves show marked steps (at $t \approx 150$ s and $t \approx 240$ s) caused by enrichment at high

load, with end-of-cycle CO ≈ 1.1 g/km — slightly above the Euro 6e limit of 1.0 g/km. This is acceptable in the present comparison because the Euro 6e limit is legally defined on the WLTC (where the vehicle is compliant), while Artemis Motorway is a much more aggressive real-driving profile; it nevertheless highlights the sensitivity of CO to high-load enrichment.

4.3 Cycle comparison

Table 2: Summary of end-of-cycle results for the three driving cycles (baseline vehicle, adaptive ECMS, no retuning). Values read from the simulation scopes.

	WLTC Class 3	Artemis Rural	Artemis Motorway 130
Duration [s]	1800	1082	1068
Final SOC [%] (start 60)	≈ 57.5	≈ 57.7	≈ 58.5
Fuel economy [l/100 km]	≈ 5.5	≈ 5.0	≈ 6.0
CO ₂ [g/km]	≈ 125	≈ 115	≈ 135
CO [g/km]	≈ 0.70	≈ 0.38	≈ 1.1
HC [g/km]	≈ 0.014	≈ 0.008	≈ 0.021
NO _x [g/km]	$\approx 8 \times 10^{-4}$	$\approx 7 \times 10^{-4}$	$\approx 1.2 \times 10^{-3}$

The ranking follows the average power demand of the cycles: the higher the mean traction power (motorway > WLTC > rural), the smaller the relative benefit of hybridization and the higher fuel consumption and emissions. The adaptive ECMS maintains charge-sustaining operation on all cycles without any retuning, which demonstrates the robustness of the adaptive equivalence factor of Eq. (4).

5 Sensitivity analysis: vehicle mass $\pm 25\%$

The vehicle mass was varied by $\pm 25\%$ around the baseline value (Table 3), keeping every other parameter and the whole ECMS calibration unchanged, and the WLTC Class 3 simulation was repeated.

Table 3: Vehicle mass cases.

	-25%	Baseline	+25%
Mass [kg]	1217	1623	2029

5.1 Results



Figure 12: WLTC Class 3, $m = 2029$ kg (+25%): overview scope.

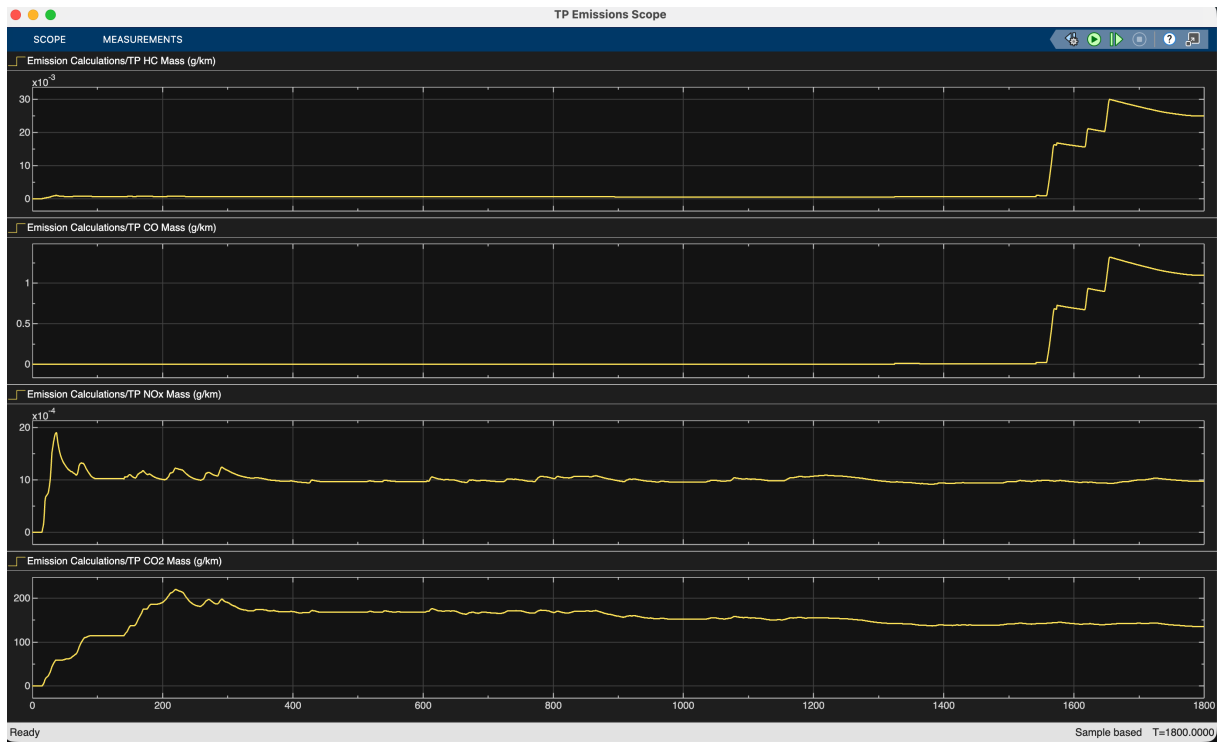


Figure 13: WLTC Class 3, $m = 2029$ kg (+25%): cumulative emissions.

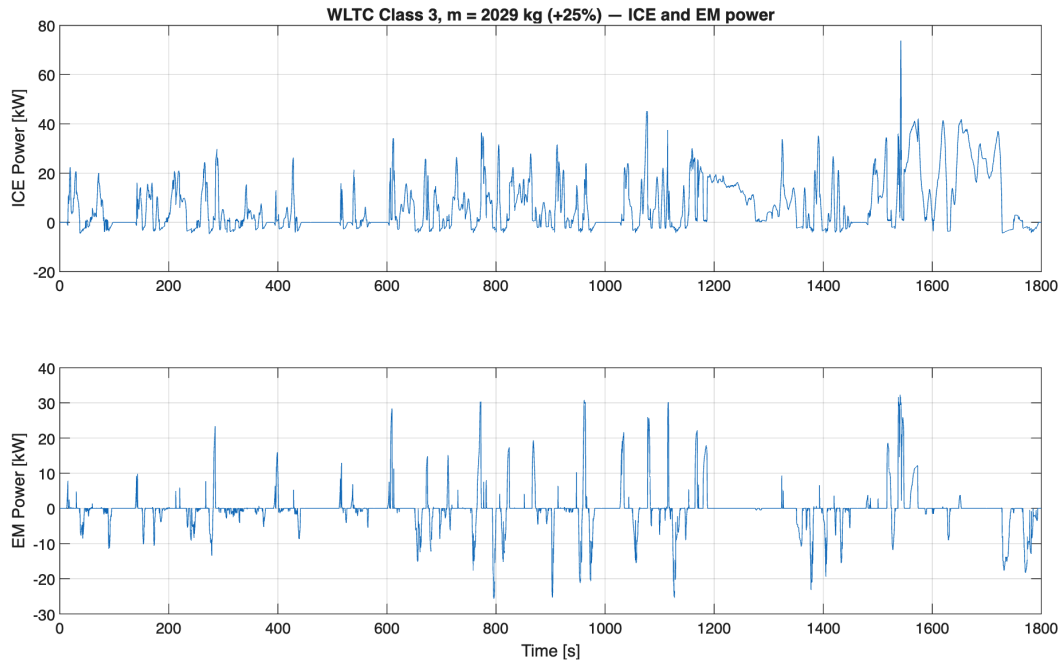


Figure 14: WLTC Class 3, $m = 2029$ kg (+25%): ICE and EM power.



Figure 15: WLTC Class 3, $m = 1217$ kg (-25%): overview scope.

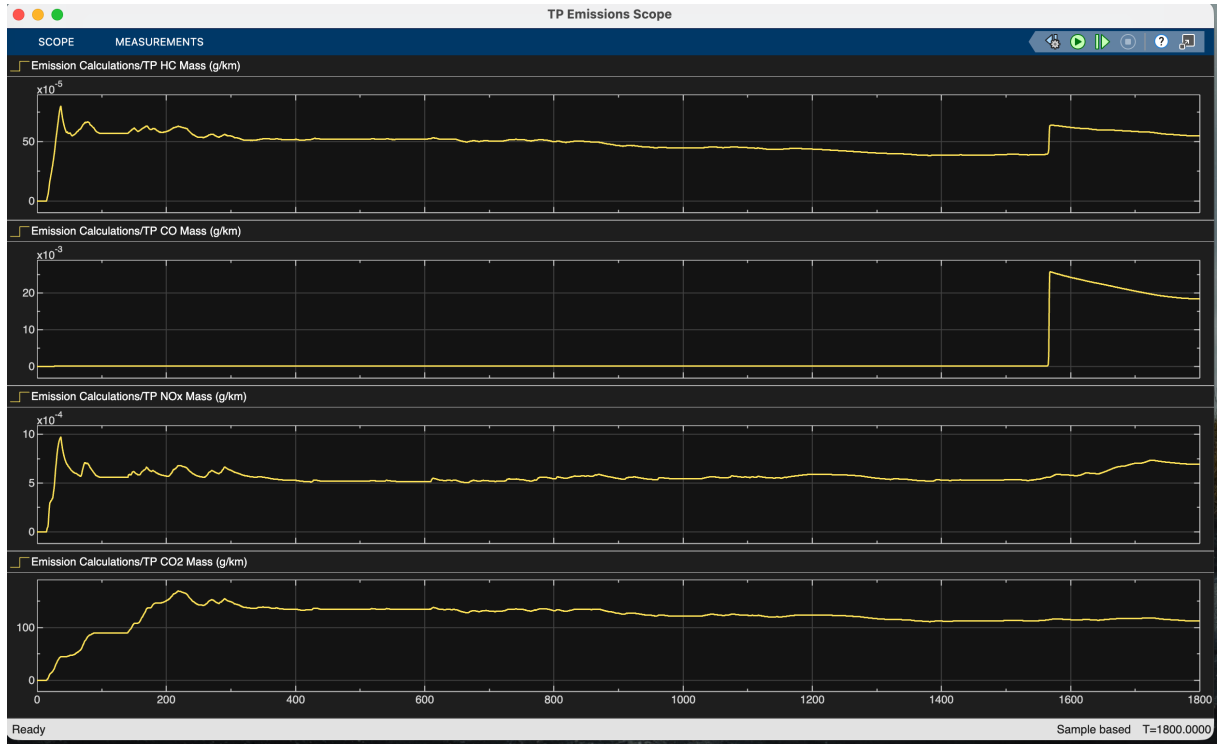


Figure 16: WLTC Class 3, $m = 1217$ kg (-25%): cumulative emissions.

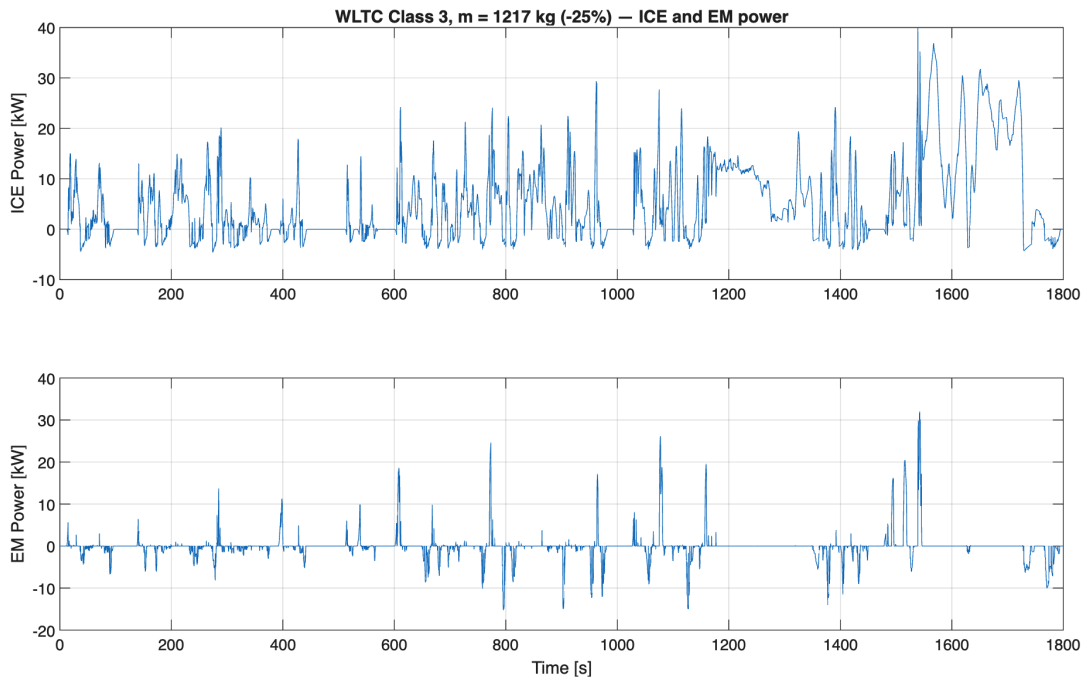


Figure 17: WLTC Class 3, $m = 1217$ kg (-25%): ICE and EM power.

The direct comparison of the instantaneous ICE and EM power for the three mass cases is shown in Figures [18](#) and [19](#).

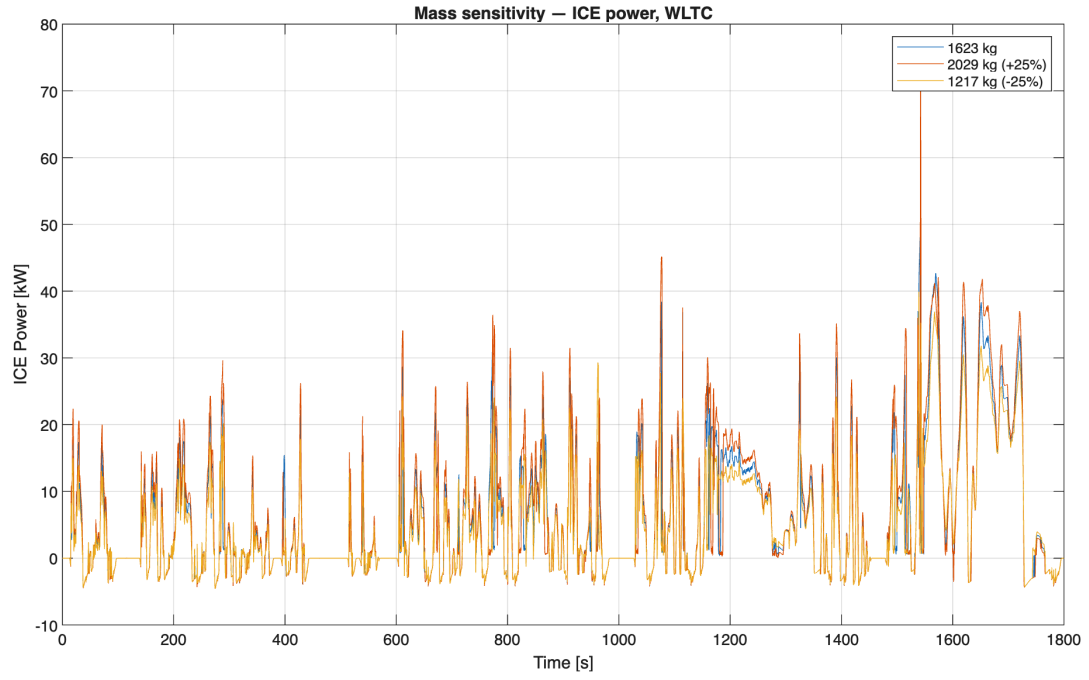


Figure 18: ICE power over the WLTC for the three mass cases.

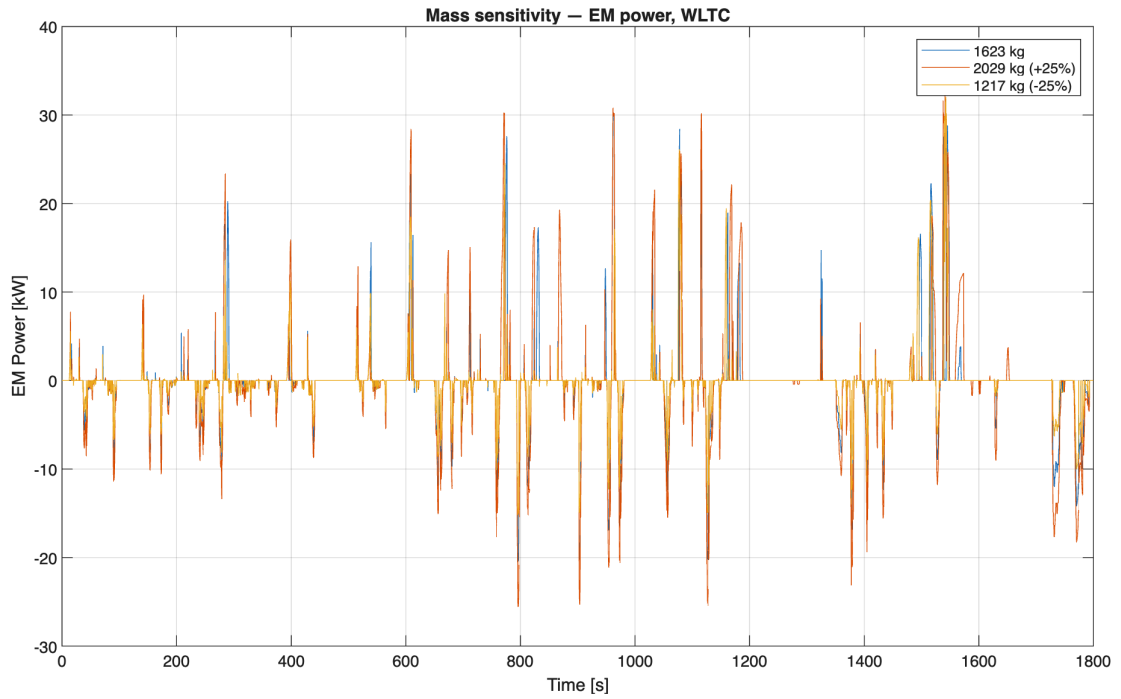


Figure 19: EM power over the WLTC for the three mass cases.

Table 4: Effect of the vehicle mass on the WLTC results (adaptive ECMS, no retuning). Values read from the simulation scopes.

	$m = 1217 \text{ kg}$	$m = 1623 \text{ kg}$	$m = 2029 \text{ kg}$
Final SOC [%]	≈ 57.5	≈ 57.5	≈ 58
Fuel economy [l/100 km]	≈ 5.0	≈ 5.5	≈ 6.5
CO ₂ [g/km]	≈ 110	≈ 125	≈ 140
CO [g/km]	≈ 0.018	≈ 0.70	≈ 1.1
HC [g/km]	$\approx 5.5 \times 10^{-4}$	≈ 0.014	≈ 0.025
NO _x [g/km]	$\approx 7 \times 10^{-4}$	$\approx 8 \times 10^{-4}$	$\approx 1.0 \times 10^{-3}$

5.2 Theoretical justification

The longitudinal dynamics equation explains the observed trends. The traction force at the wheels is

$$F_t = \underbrace{ma}_{\text{inertia}} + \underbrace{mgc_r \cos \alpha}_{\text{rolling}} + \underbrace{\frac{1}{2}\rho c_x A v^2}_{\text{aerodynamic}} + \underbrace{mg \sin \alpha}_{\text{grade}} \quad (6)$$

Two of the four terms (inertia and rolling resistance) are directly proportional to the mass. On a transient-rich cycle such as the WLTC the inertia term dominates during accelerations, therefore:

- **+25% mass:** the traction power demand grows almost proportionally during transients; the ICE operates longer and at higher loads (Figure 18), enrichment phases become more frequent (visible as the larger HC/CO steps in Figure 13), and fuel consumption and CO₂ rise by roughly +15–20%. The kinetic energy available at each braking, $E_k = \frac{1}{2}mv^2$, also grows linearly with m , so the regenerative (negative) EM power peaks increase (Figure 19); however, since the P1 machine and the battery power limits are unchanged, only part of the additional braking energy can be recovered and the net efficiency worsens.
- **–25% mass:** the demanded power decreases; the engine operates more often at light load, but the total energy demand reduction prevails, and both fuel consumption ($\approx 5.01/100 \text{ km}$) and CO₂ ($\approx 110 \text{ g/km}$) improve. The dramatic reduction of CO and HC (two orders of magnitude for HC) indicates that with the reduced load the engine practically never enters the enrichment region during the extra-high phase.
- The adaptive ECMS keeps the SOC balanced in all cases (final SOC ≈ 57.5 –58%) without retuning, confirming the effectiveness of the SOC feedback on the equivalence factor. With a *constant* EF tuned for 1623 kg, the $\pm 25\%$ mass cases would instead have ended with a visible SOC drift, because the optimal equivalence factor depends on the cycle energy balance.

As a first-order check, differentiating the cycle fuel energy with respect to mass at constant efficiency gives $\partial \dot{m}_f / \partial m \propto (a + gc_r)v$, i.e. a roughly linear dependence of consumption on mass on a transient cycle: the simulated $\pm 25\%$ mass change produced approximately $\mp 9\%$ / $+18\%$ changes in fuel consumption, consistent with the fact that only the inertia and rolling terms of Eq. (6) scale with m while the aerodynamic term does not.

6 Conclusions

A P1 hybrid electric vehicle with an ICE torque multiplier of 1.02 was simulated in the MathWorks HEV P1 virtual vehicle environment under ECMS control. The main findings are:

1. A constant equivalence factor $s = 3.23$ achieves SOC balance on the WLTC Class 3; the adaptive ECMS reproduces charge-sustaining operation on every tested condition without retuning.
2. Over the WLTC the vehicle achieves ≈ 5.5 l/100 km and ≈ 125 g/km of CO₂; HC, CO and NO_x are compliant with Euro 6e with wide margins, while the CO₂ value remains above the 2025 EU fleet target of 93.6 g/km, showing the limits of mild (P1) hybridization for a vehicle of this class.
3. The ECMS shifts the engine operating points towards the low-BSFC region of the map and prevents steady light-load operation; the P1 electric machine works in a narrow low-torque band (assist and regeneration) at engine speed, with intrinsically limited energy throughput.
4. Cycle comparison (Artemis Rural / WLTC / Artemis Motorway) shows fuel consumption and emissions increasing with the mean power demand of the cycle; on the motorway cycle the hybrid benefit almost vanishes and high-load enrichment raises CO close to (and beyond) the Euro 6e number.
5. The $\pm 25\%$ vehicle mass variation changes fuel consumption and CO₂ roughly linearly (inertia- and rolling-dominated), modifies the ICE operating point distribution accordingly, and scales the recoverable braking energy; the adaptive ECMS absorbs the parameter change while keeping the SOC balanced.

A Post-processing scripts

The operating-point and power plots were generated in MATLAB from the logged signals (logsout: EngSpdOut, EngTrq, EMSpd, EMTrq) and from the maps stored in the ECMS block (f_tbrake_n_bpt, f_tbrake_t_bpt, f_fuel, f_mtr_w_bpt, f_mtr_t_bpt, f_mtr_eta), resolved with slResolve. The BSFC map was computed as $\text{BSFC} = 3.6 \times 10^9 \dot{m}_f / (T \omega)$ [g/kWh] with the low-power region masked, and the EM efficiency map was mirrored to negative torque for generator operation. The complete scripts (bsfc.m, ice_over_bsfc.m, ice_em_power.m, results.m) are stored with the result images.